

NLPTEL

Natural Language Processing

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Natural Language Processing (NLP)

Natural language processing (NLP) is technology that allows computers to interpret, manipulate, and comprehend human language.

NLP is used by many applications such as text translation, text summarization, speech to text conversion, chatbots etc.

- ✓ Conventional NLP refers to the rule-based and statistical approaches to processing human language.
- ✓ Deep learning-based NLP refers to approaches that use neural networks—particularly deep architectures like recurrent, convolutional, and transformer models—to process and understand human language.

NLP Tasks

Sentiment Analysis

Text Classification

Machine Translation

Text Summarization

Question answering

Text Entailment

Chatbots

Named Entity Recognition

Language Modeling

Sentiment Analysis

Sentence

“I absolutely love this phone!”

“This restaurant is terrible.”

“The movie was okay, nothing special.”

“I have never tasted a pizza like this”

Sentiment

Positive

Negative

Neutral

.....

Text Classification

Text

Label

“Win a free iPhone now!
Click here!”

Task
Spam Detection

Spam

Your order has been shipped.”

Task
Spam Detection

Not Spam

“The stock market closed higher
today after...”

Task
Topic Classification

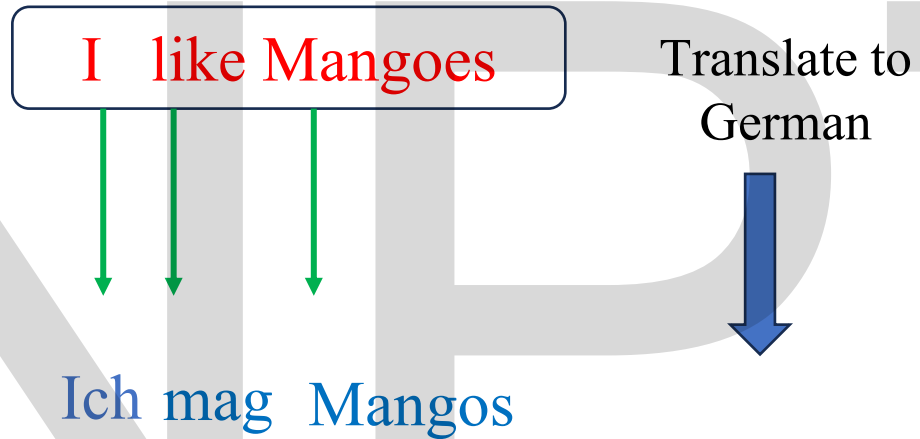
Business/Finance

Cristiano Ronaldo scores a
hat-trick...”

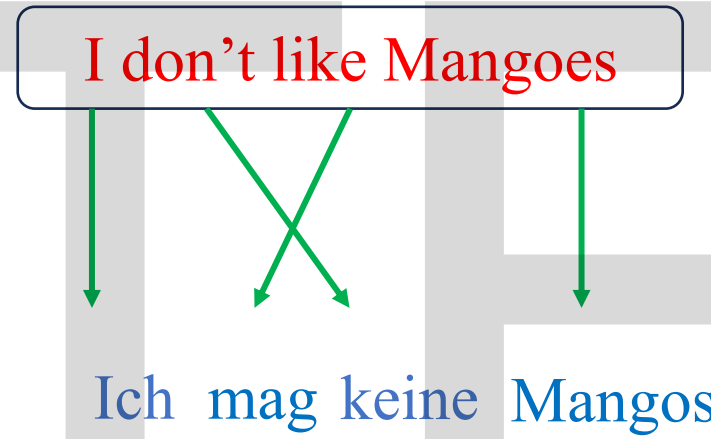
Task
Topic Classification

Sports

Machine Translation



Sequence of words in the input and output sentence are same



Sequence of words in the input and output sentence are different

Text Summarization

Original Text

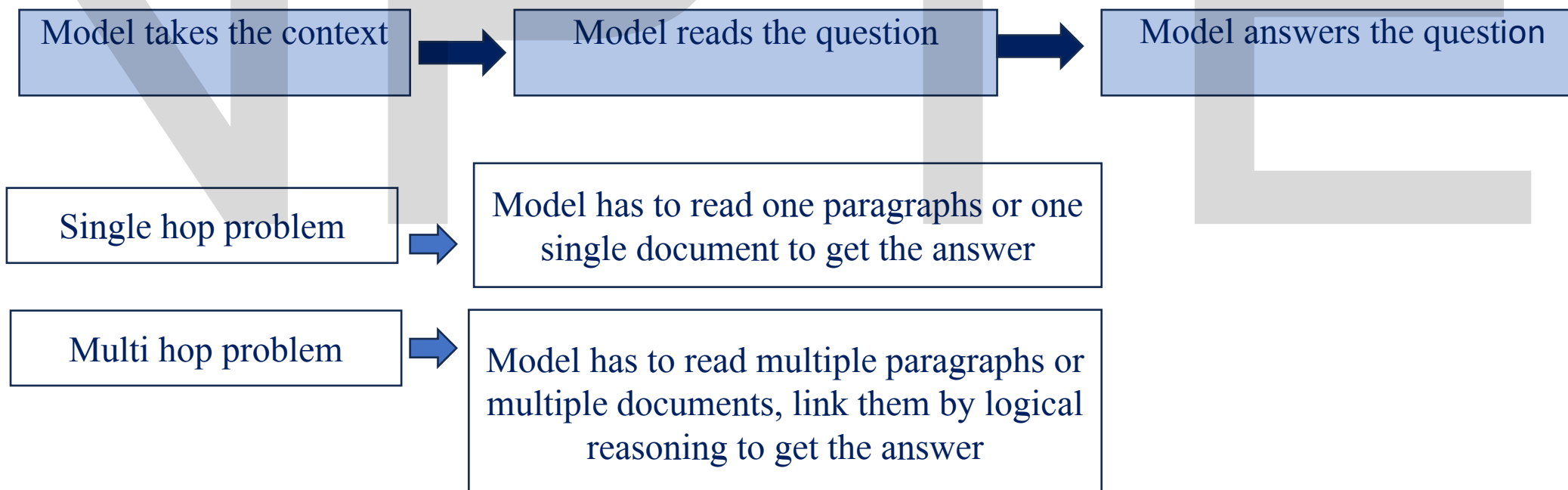
“NASA’s James Webb Space Telescope captured detailed images of a distant exoplanet’s atmosphere, revealing evidence of water vapor, clouds, and possible signs of life-supporting conditions. Scientists say this is a major breakthrough in the search for habitable worlds.

Summary

“NASA’s James Webb Telescope found water vapor and clouds on a distant exoplanet, a big step in the search for habitable worlds.”

Question Answering

Question	Answer
“Who was the first woman to win a Nobel Prize?”	Marie Curie



Text Entailment

Two sentences will be given- Model have to tell if these sentences contradict, entail or neutral

Example:

A black race car starts at the front of the crowd

A black race car starts at the front of a lonely road

A man is riding a bicycle in the park.

A person is cycling outdoors.

He is playing cricket with the kids

The kids are enjoying ice cream

Contradicts

Entails

Neutral

Chatbot

Customer Support Chatbot

User

- Hi, I ordered a phone last week. Can you check its delivery status?
- It's #54732.
- Great. Can I change the delivery address?

Chatbot

- ✓ Sure! Could you provide your order ID, please?
- ✓ Thanks! Let me check... Your phone was shipped yesterday and is expected to arrive on Friday.
- ✓ Yes, you can. Please click this link to update your address before 6 PM today.

Named Entity Recognition (NER)

NER detects **real-world objects** mentioned in text and labels them with their entity type.

Sourav Ganguly served as the captain of Indian Cricket team and he is from Kolkata

NER Output:

- Sourav Ganguly → **PERSON**
- Indian Cricket team → **ORGANIZATION**
- Kolkata → **LOCATION**

Other NER includes

- **TIME** – specific times
- **DATE** – date expressions
- **MONEY** – monetary values
- **PERCENT** – percentages
- **FACILITY** – buildings, airports
- **EVENT** – historical or social events

Language Modeling

A language model in natural language processing (NLP) is a statistical or machine learning model that is used to predict the next word in a sequence given the previous words.

Input

“I’m going to the”

“The sun rises in the”

“Artificial intelligence is transforming
the.....”

Predicted next word

store / park / market

east

world by automating tasks and
improving efficiency.

Sentence completion

Once upon a time in a magical
forest...”(prompt)

Generated Text:

“...there lived a tiny dragon who loved to paint rainbows
in the sky.”

Challenges in NLP

Discrete Nature of
Text data

Sky and stars
Red and pink



Some relation is there among these pair of words
but words doesn't tell anything about the similarity

Evolving nature of
Natural Language

New words gets added to the dictionary (quaratene) time to time

Ambiguity/Vagueness

Never tasted a pizza like this before

Capturing long range
dependency

Animal did not cross the road as it was (wide/tired)

NPTEL

Summary

NLP TELL

Next Session

Text Preprocessing and Text Representations